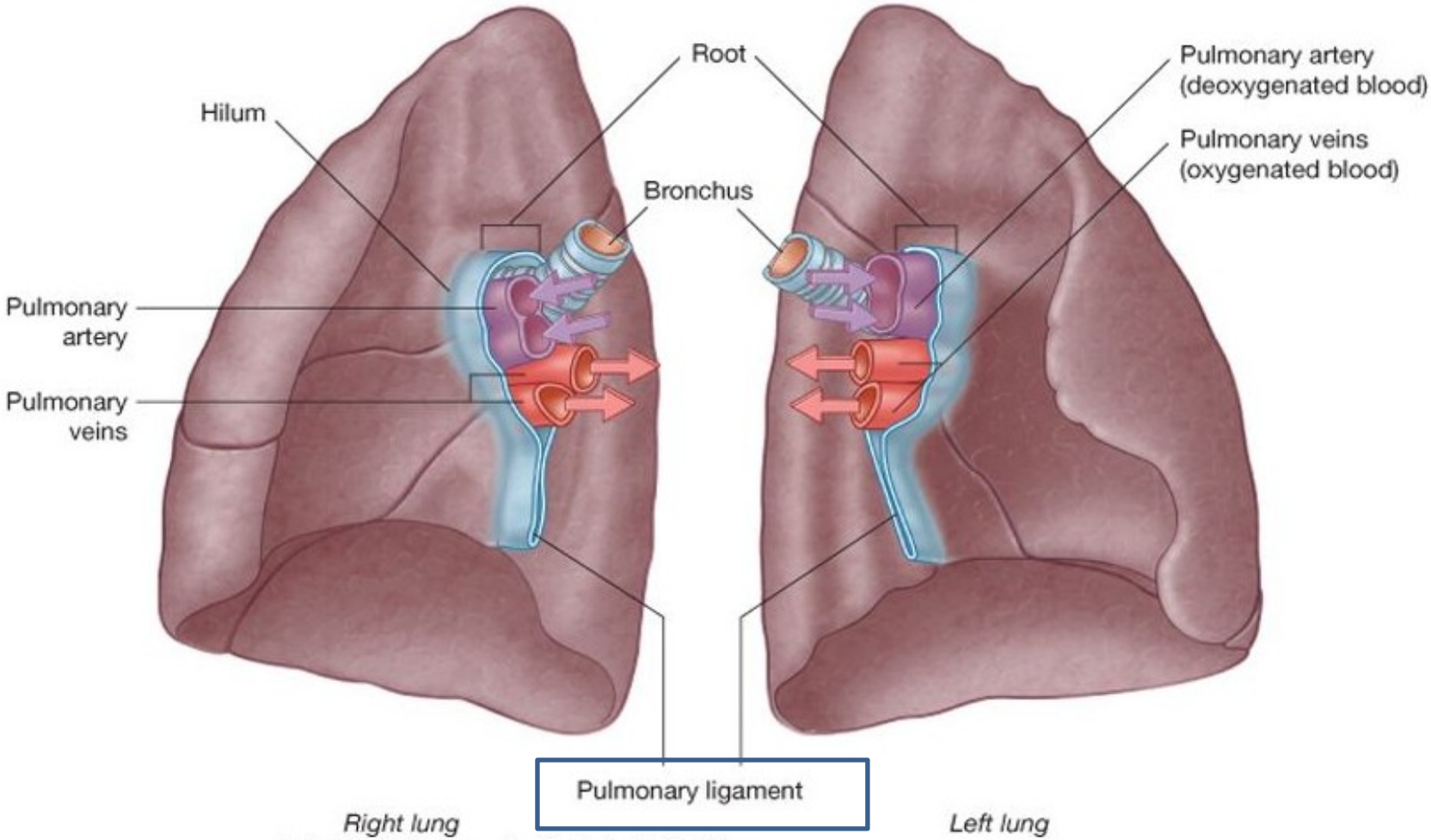
LUNGS & BRONCHOPULMONARY SEGMENTS

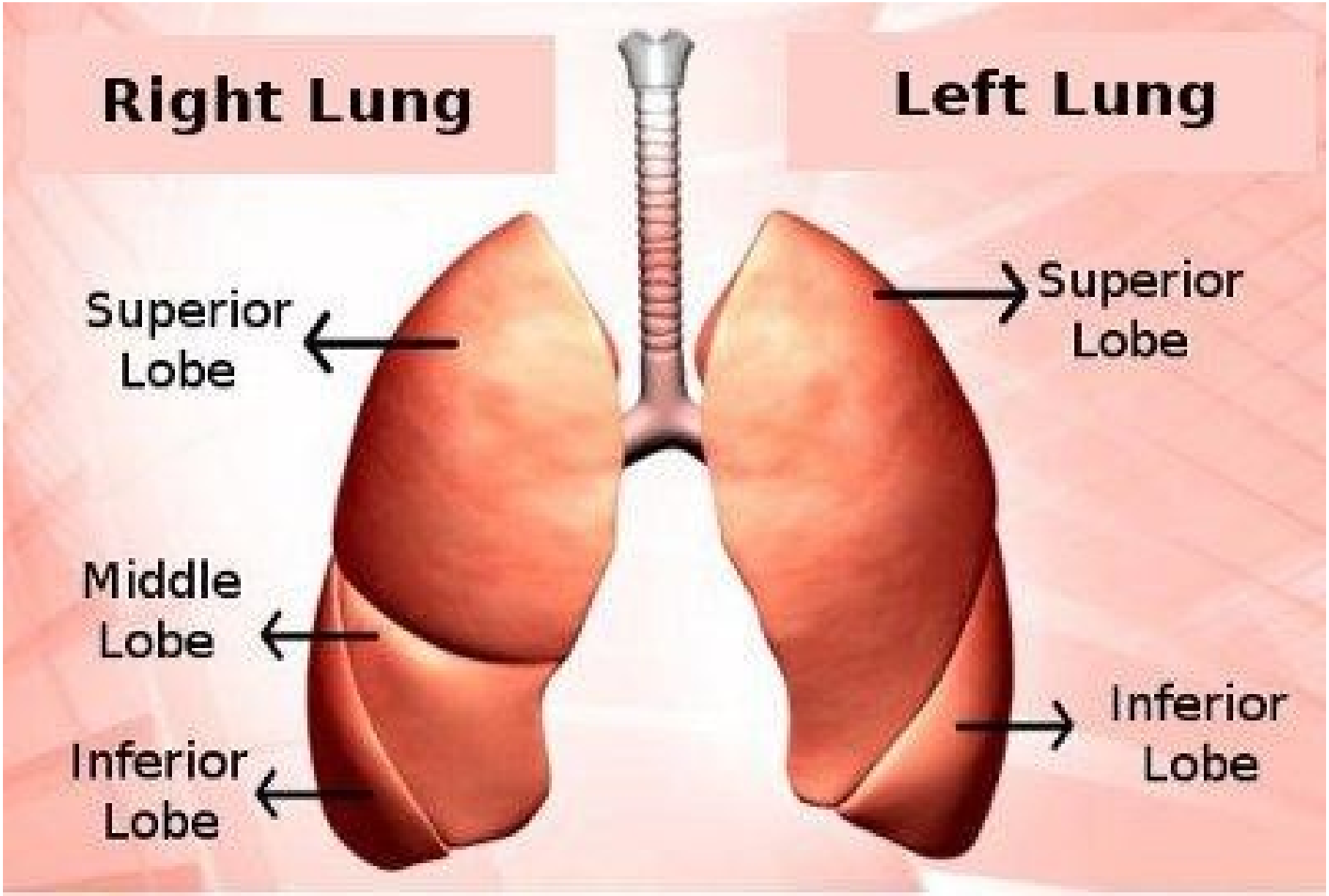


Drake: Gray's Anatomy for Students, 2nd Edition.
Copyright © 2009 by Churchill Livingstone, an imprint of Elsevier, Inc. All rights reserved.

Figure 3.40 Roots and hila of the lungs.

LUNGS

- Lungs are the vital organs of respiration.
- Function: Oxygenation of blood by bringing inspired air in close relation with the venous blood in the pulmonary capillaries
- Healthy lungs in living people are normally light, soft and spongy.
- They fully occupy the pulmonary cavities.
- They are elastic and have tendency to recoil.
- The Lungs are separated from each other by mediastinum and its contents.

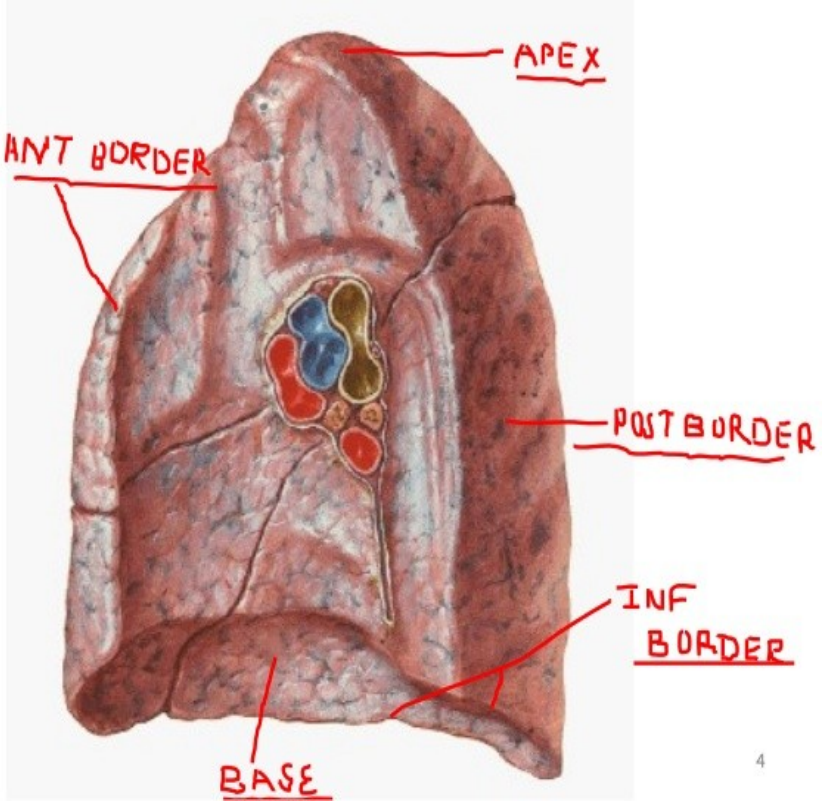

Right Lung
Left Lung
Superior Lobe
Middle Lobe
Inferior Lobe
Superior Lobe
Inferior Lobe

- Right Lung is larger and heavier than the left but it is shorter and wider because the right dome of diaphragm is higher and the pericardium and heart bulge more to the left.



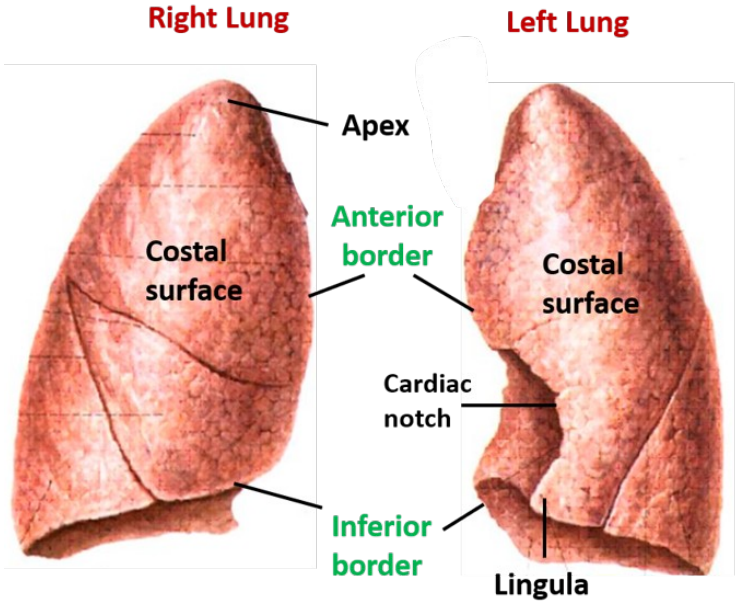
BORDERS OF THE LUNG

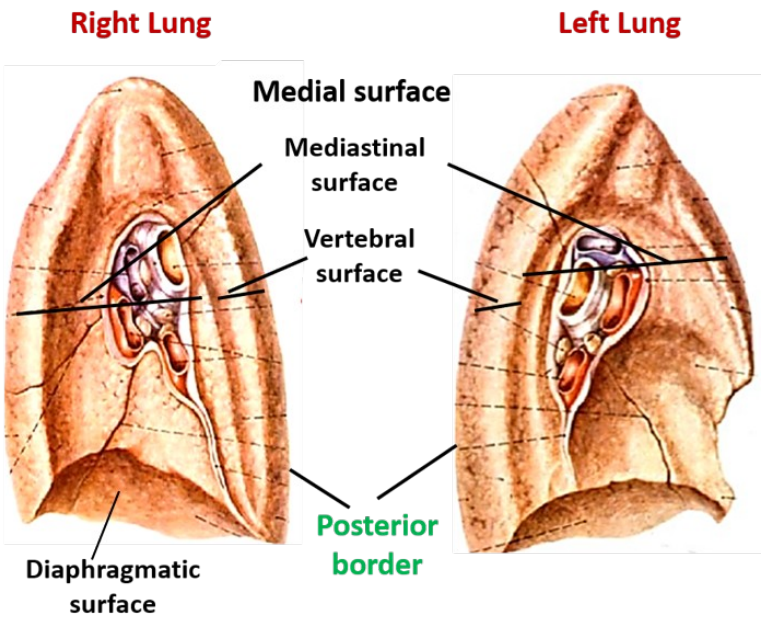
- ANTERIOR BORDER
- POSTERIOR BORDER
- INFERIOR BORDER



SURFACES OF THE LUNG

- Costal surface
- Diaphragmatic surface
- Mediastinal surface





Mediastinal Surface of the Left Lung

1- ***Cardiac impression,*** related to left ventricle.

2- Arch of the aorta.

3- Descending aorta.

4- Left common carotid artery.

5- Left subclavian artery.

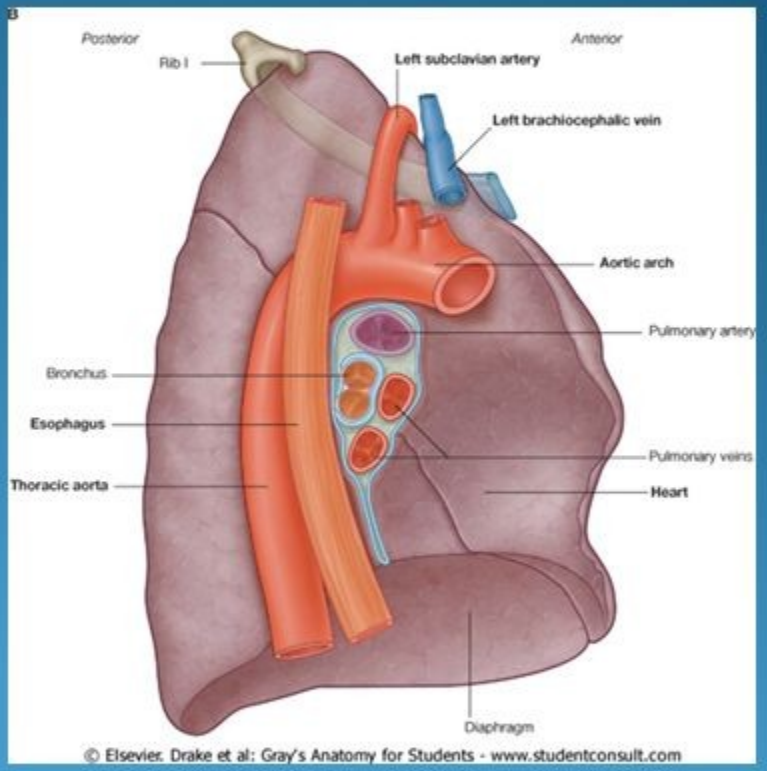
6- Oesophagus (two sites).

7- Thoracic duct.

8- Left vagus nerve.

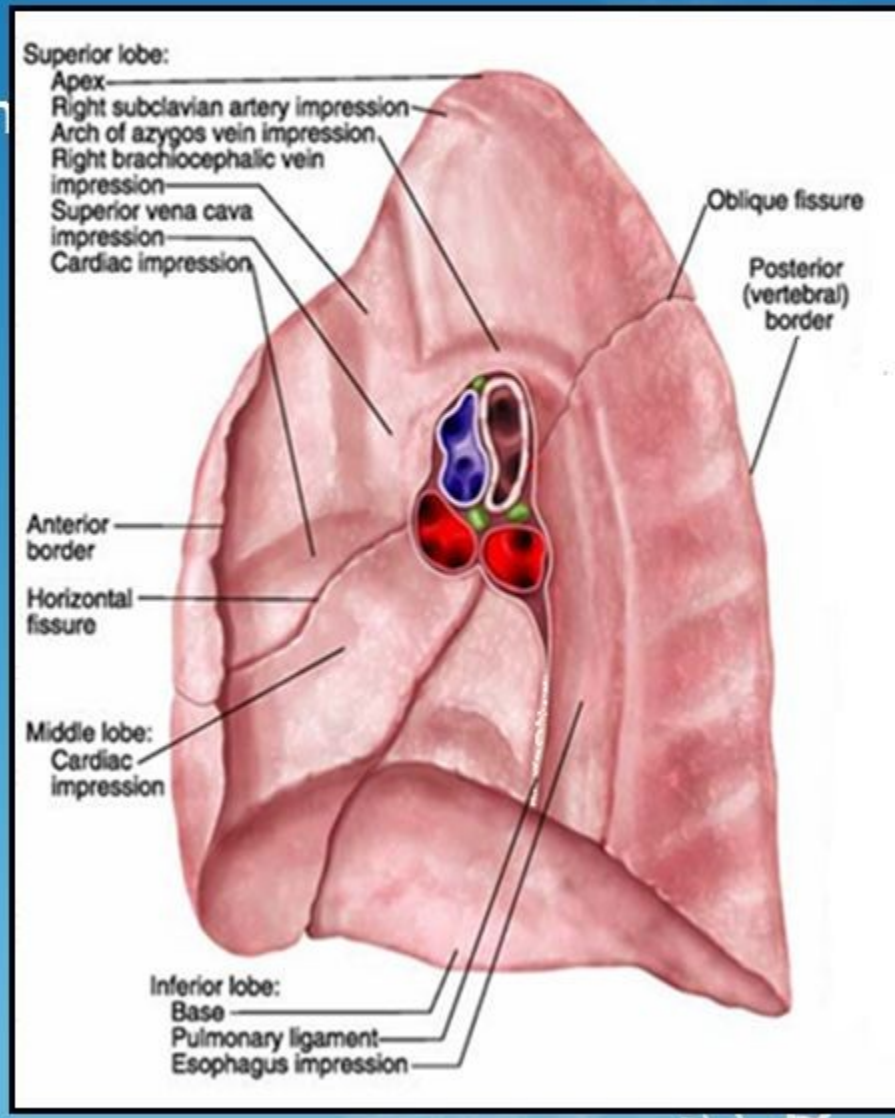
9- Left phrenic nerve.

10- Pulmonary trunk and thymus.



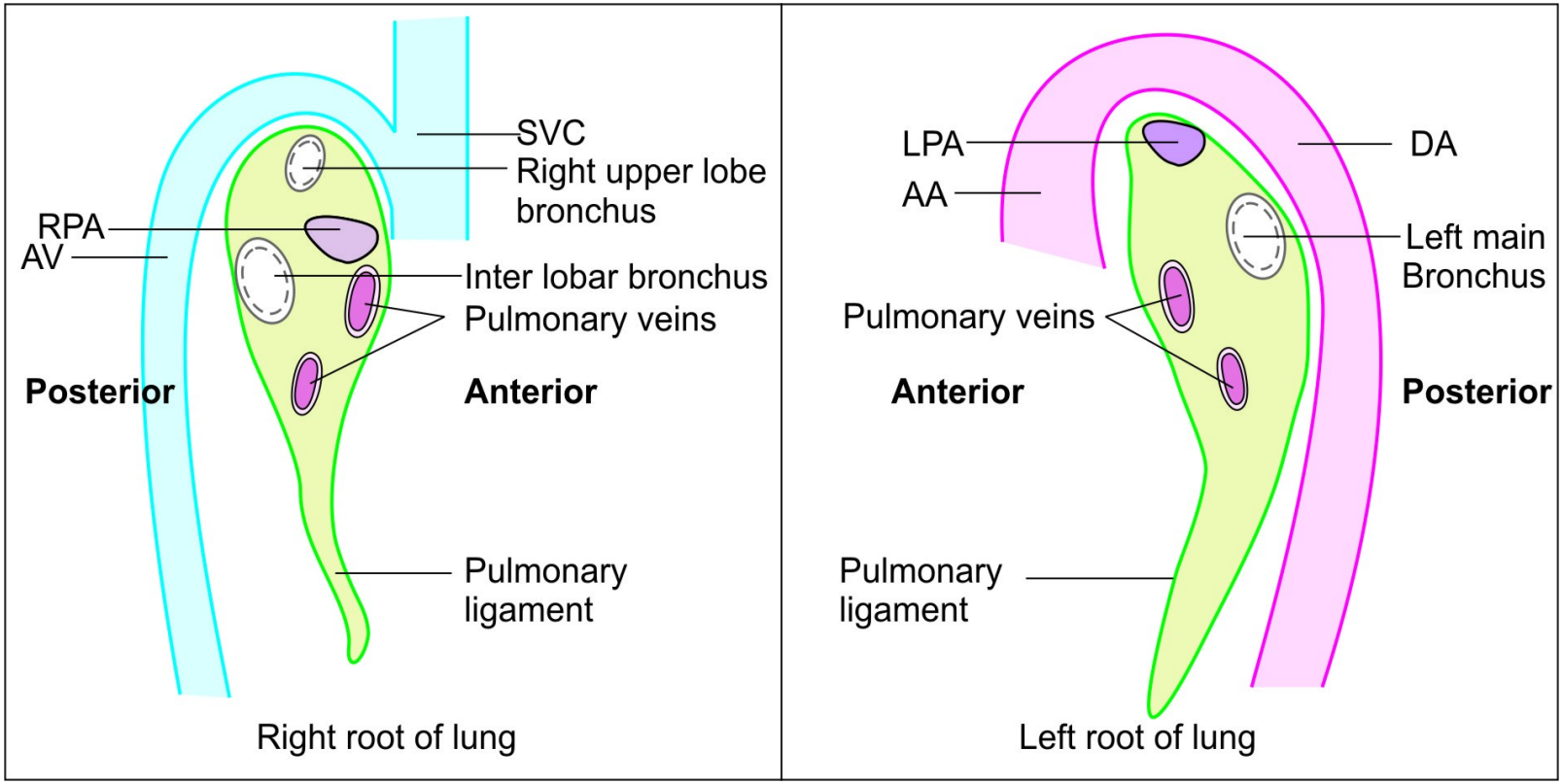
Mediastinal Surface of the Right Lung

- Cardiac impression (Right atriun
- Superior vena cava.
- Inferior vena cava.
- Right subclavian artery.
- Right brachiocephalic vein
- Azygos vein.
- Esophagus.
- Trachea.
- Thymus



ROOT OF THE LUNG

Structures entering or leaving the lung
Bronchi, pulmonary artery, pulmonary veins, lymph vessels, bronchial vessels and nerves
Surrounded by tubular sheath of pleura
the root of the lung lies opposite to 5,6,7 thoracic vertebrae



ARRANGEMENT OF STRUCTURES IN ROOT

- **FROM ANTERIOR TO POSTERIOR:**

Sup. Pulmonary vien

Pulmonary artery

Bronchus

FROM ABOVE DOWNWORDS:

RIGHT SIDE:

Eparterial bronchus

Pulmonary artery

Hyparterial bronchus

Inferior pulmonary vein

LEFT SIDE:

Pulmonary artery

Bronchus

Inferior pulmonary vein

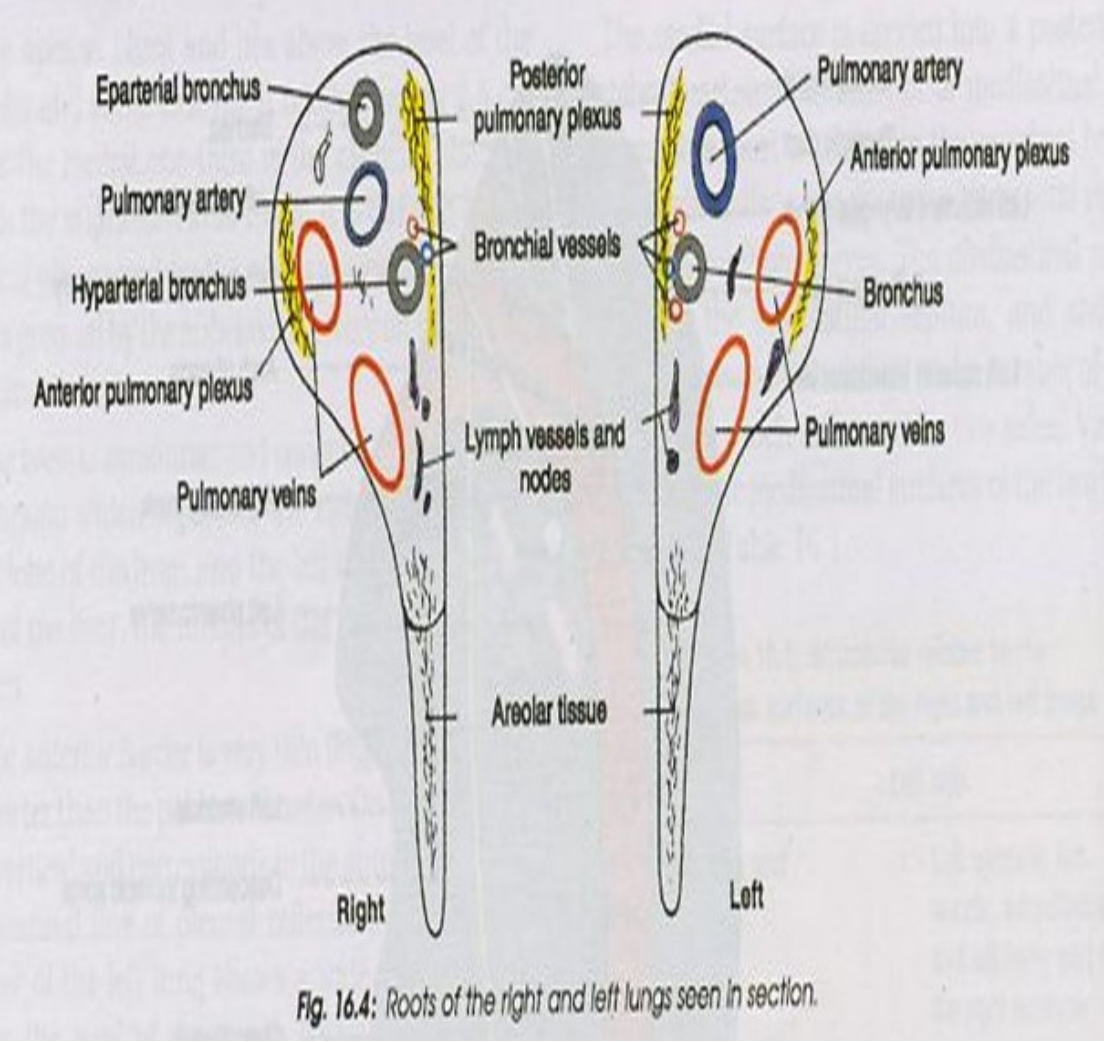
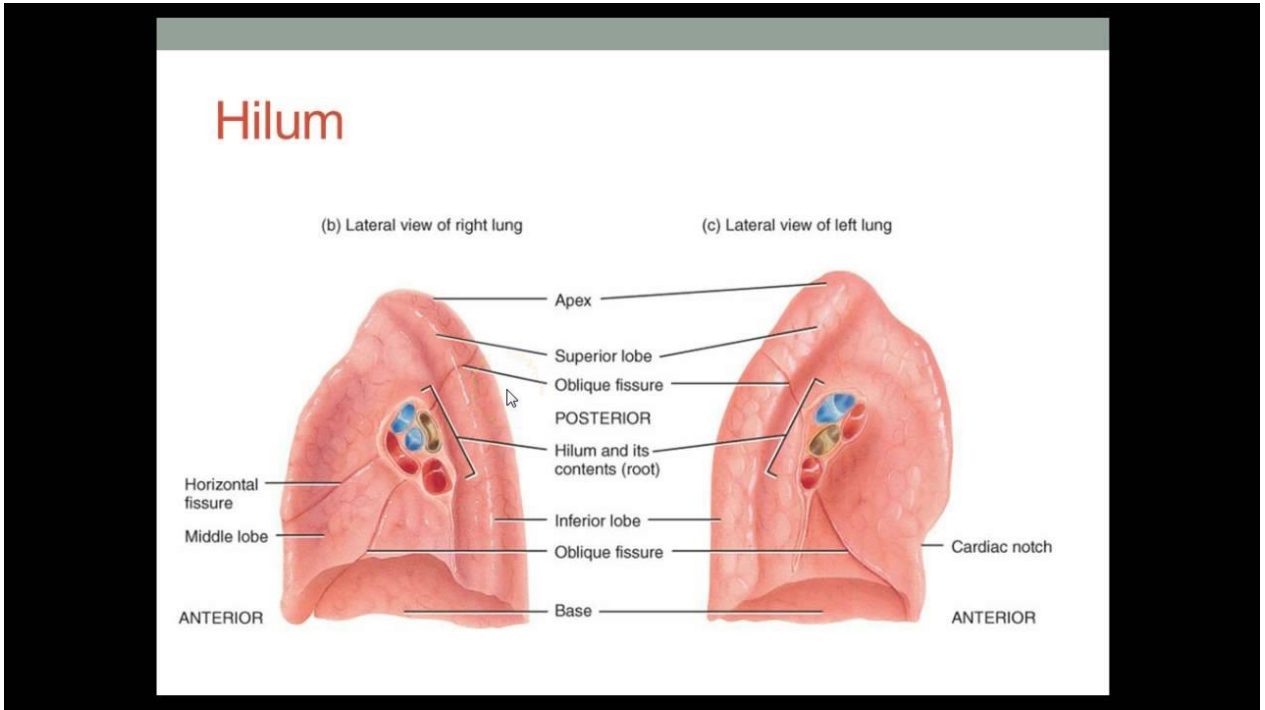


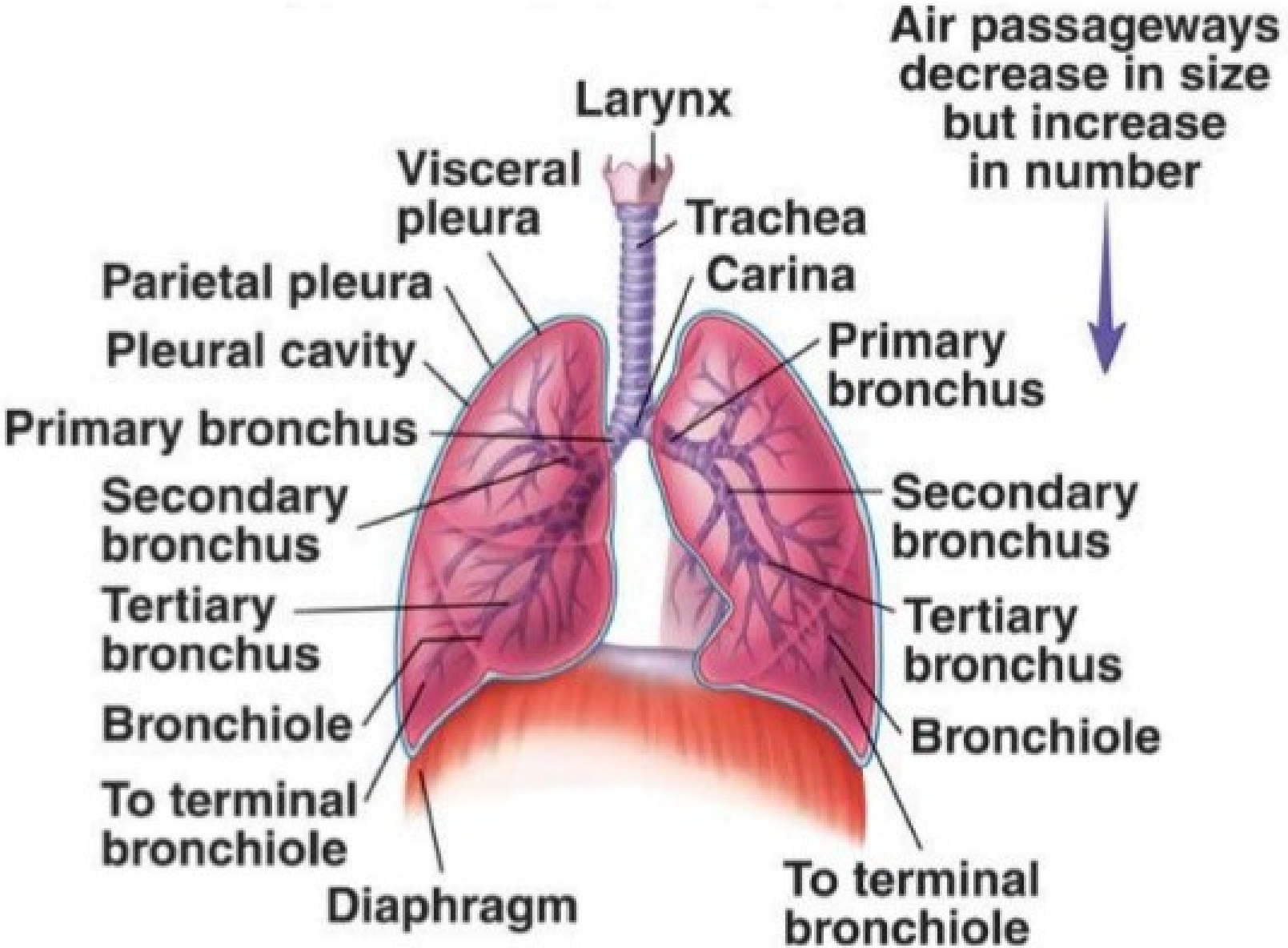
Fig. 16.4: Roots of the right and left lungs seen in section.

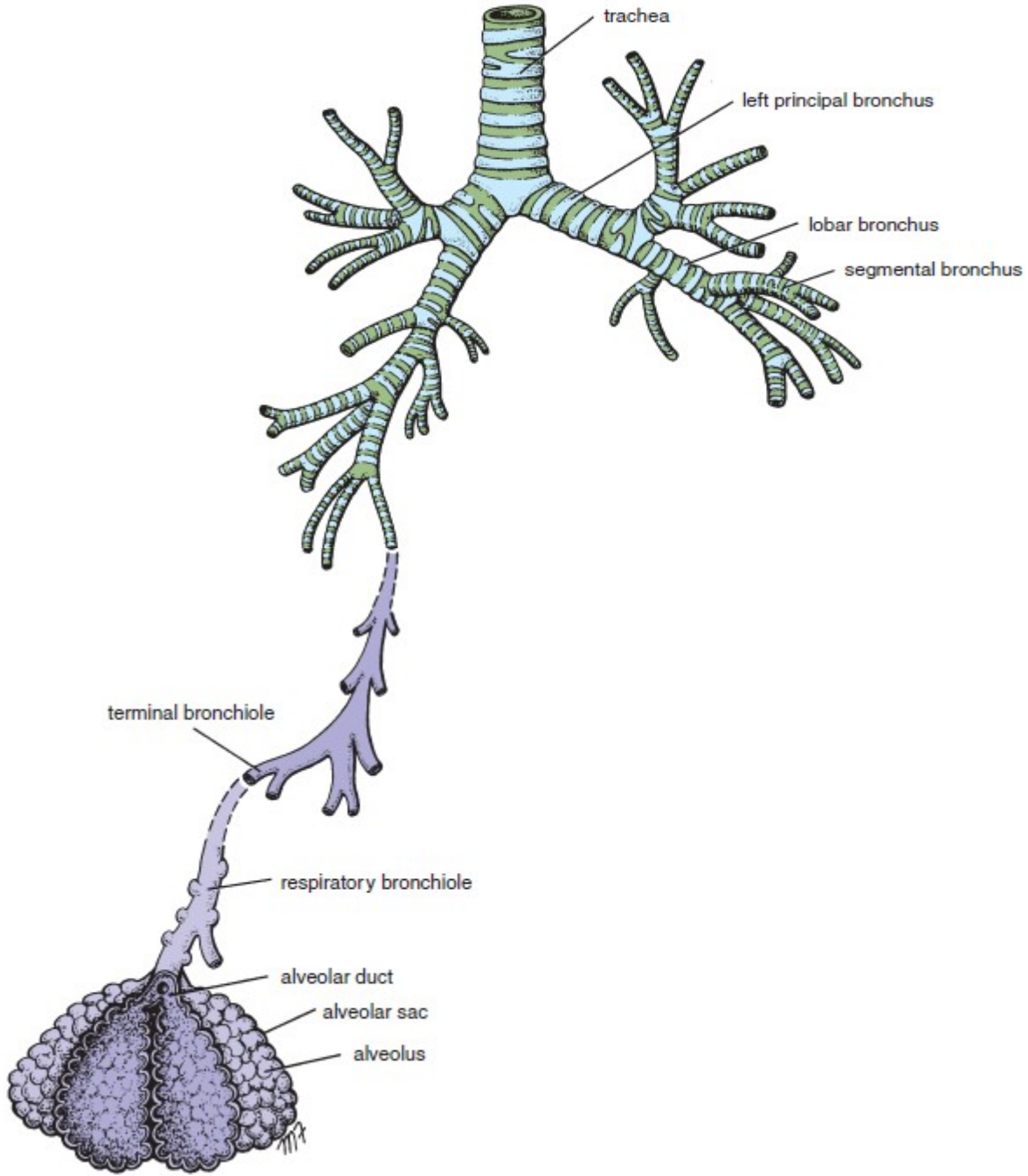
HILUM OF THE LUNG

- Also called doorway to the lung.
- It’s a wedge shape area on the mediastinal surface of Lung.
- Through it, the structures forming the root of lung, passes through.

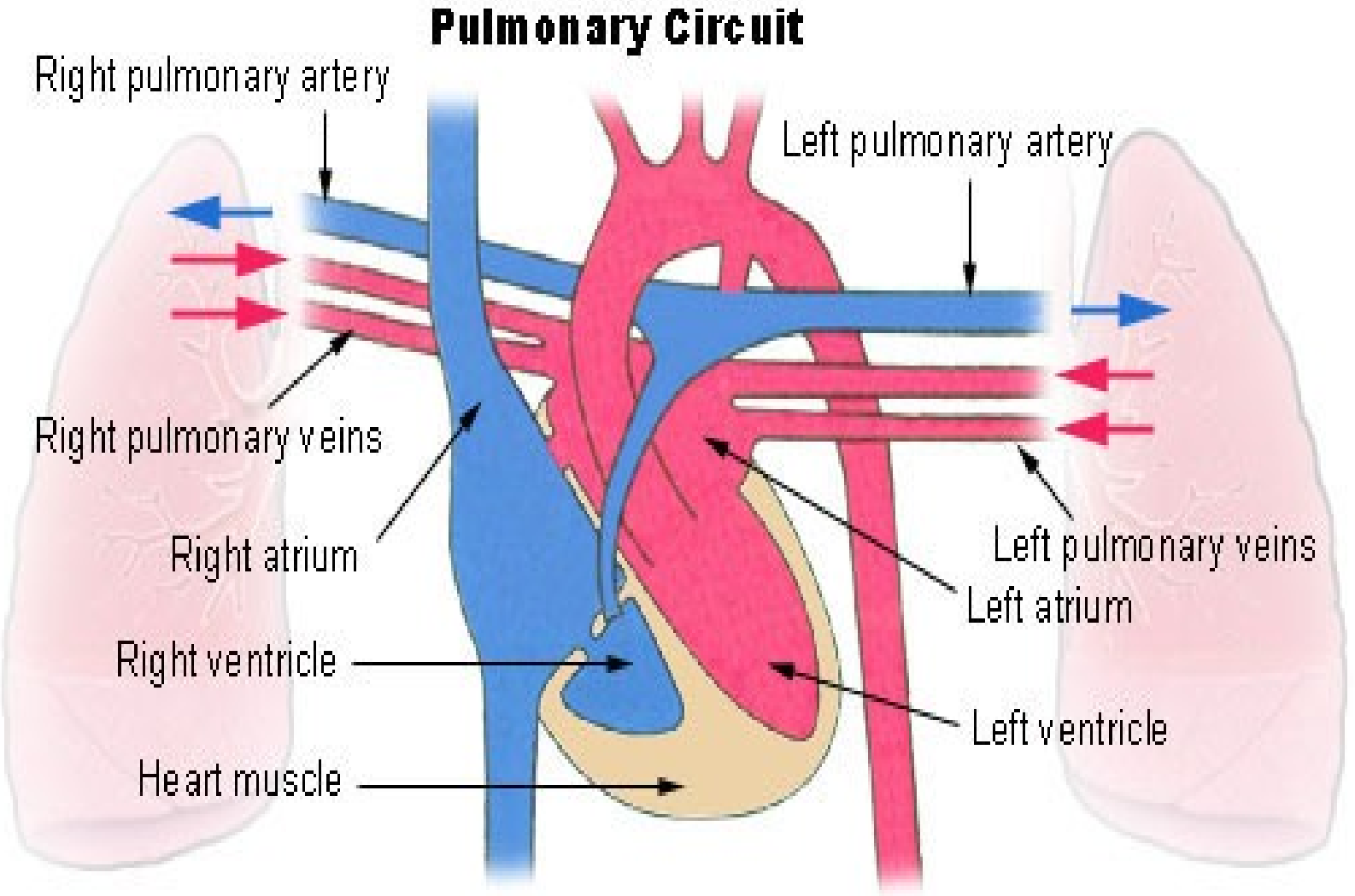


Bronchial Tree

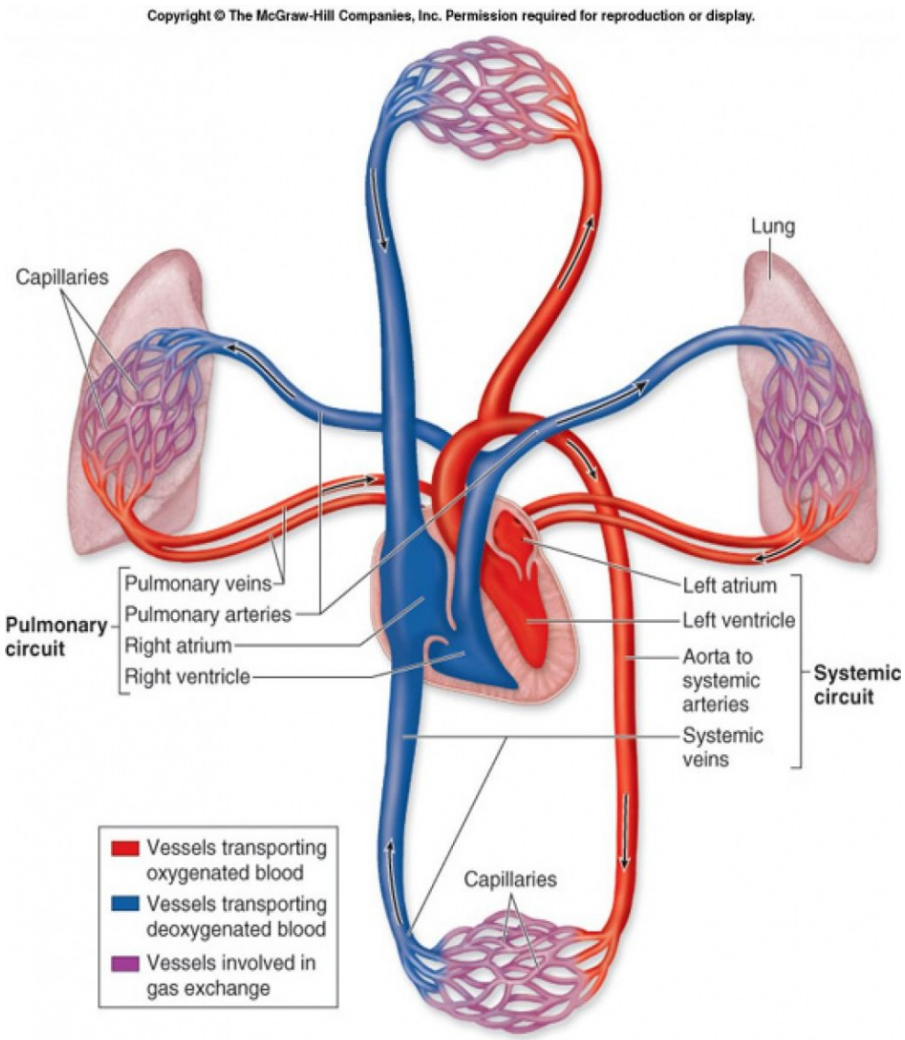



trachea
left principal bronchus
lobar bronchus
segmental bronchus
terminal bronchiole
respiratory bronchiole
alveolar duct
alveolar sac
alveolus

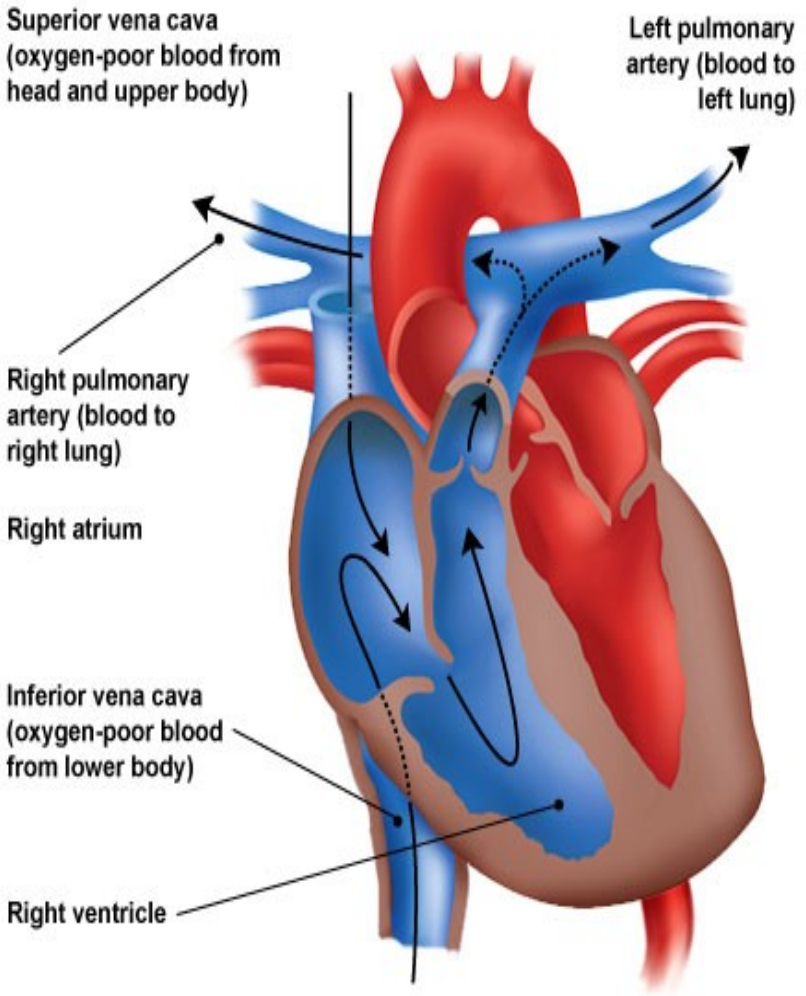
BLOOD SUPPLY OF THE LUNG



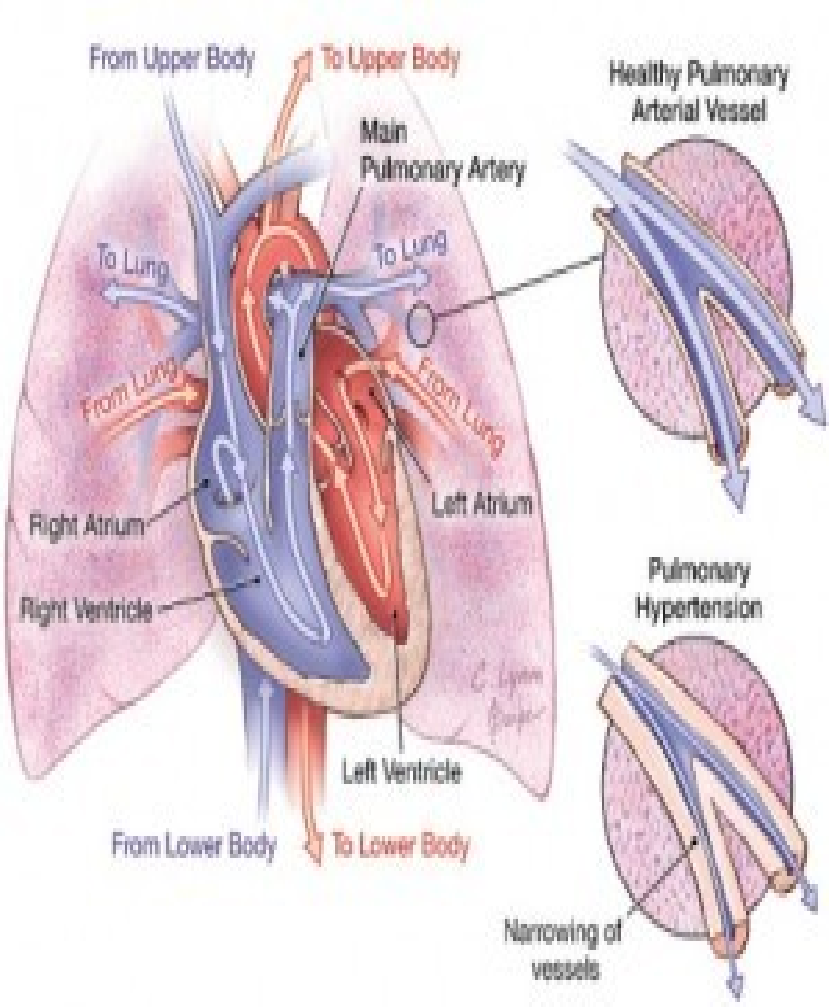
- Each lung has a pulmonary artery supplying blood to it.
- R & L pulmonary arteries arise from pulmonary trunk at the level of sternal angle.
- They carry low oxygenated i.e. venous blood to lungs



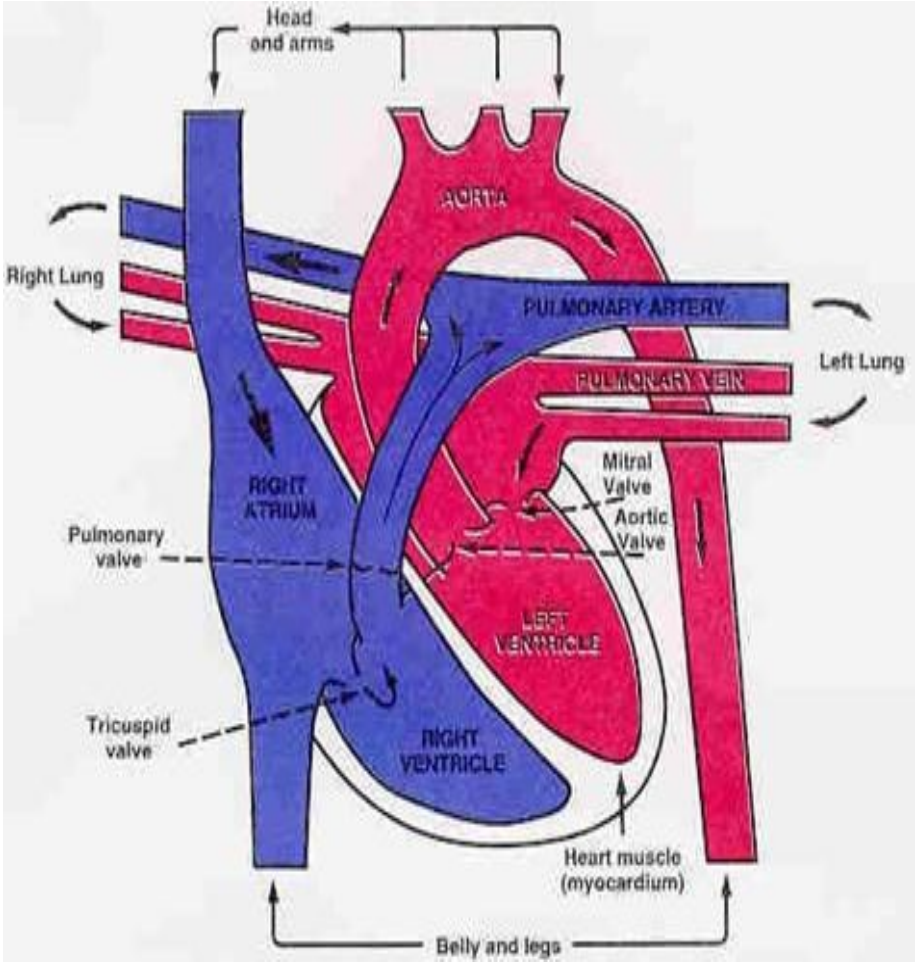
- Each pulmonary artery divide : secondary lobar As: tertiary segmental As.
- Arteries and bronchi are paired in the Lungs.
- Lobar bronchi & secondary lobar arteries serves each lobe.



- Tertiary/segmental bronchi paired with T/Segmental arteries supplies broncho-pulmonary segments.
- Pulmonary veins are intersegmental.
- Two pulmonary veins: superior & inferior on each side: carry oxygen rich blood: to the left atrium.

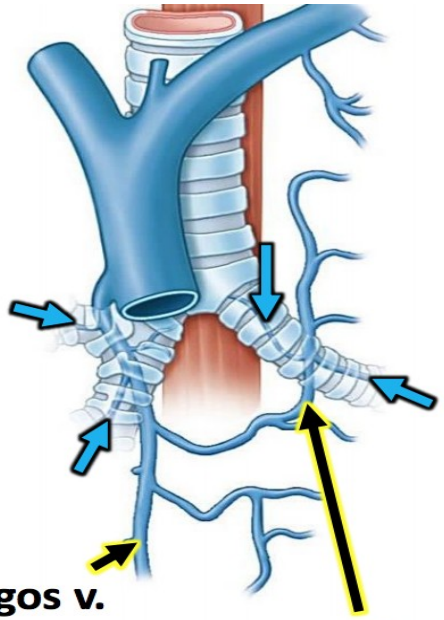


- The Bronchial arteries supply blood for nutrition to: structures making the root, supporting tissues and visceral pleura.
- There is one right and two left bronchial arteries.



VENOUS DRAINAGE

- Carried by bronchial veins
- 2 in number on each side.
- Right bronchial vein drains into azygos vein
- Left bronchial vein drains into hemiazygos vein



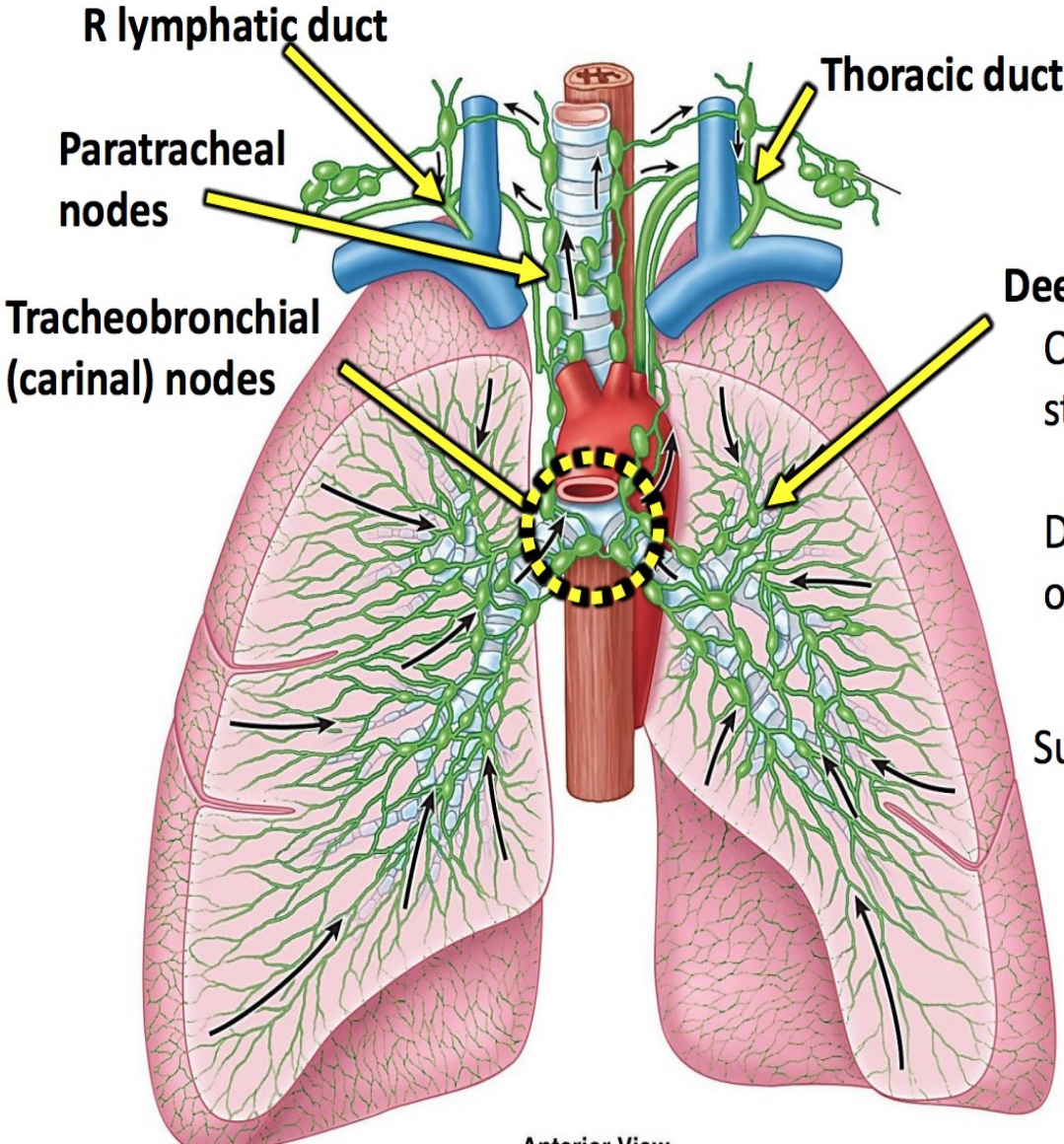
Bronchial vv. ➡
Mainly drain airway walls & stroma near root of lung

Right bronchial vv. → azygos v.

Left bronchial vv. → accessory hemiazygos v.

Veins of thoracic wall
Drain parietal pleura

LYMPHATIC DRAINAGE OF THE LUNGS



Deep plexus

Originates in walls of airways & surrounding stroma

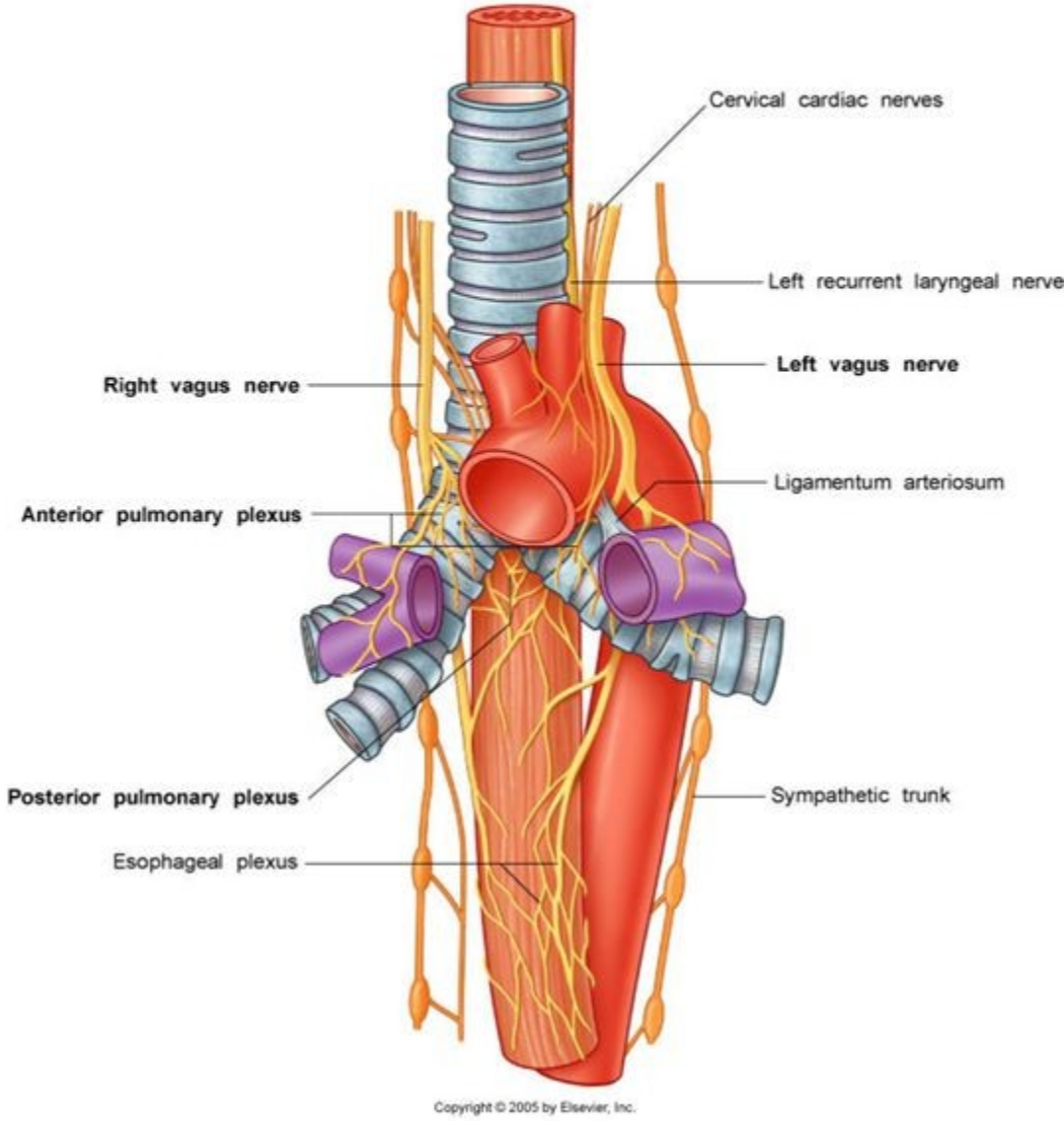
Drains lymph from structures that form root of lung

Superficial & deep lymph drain sequentially to:

Hilar (bronchopulmonary) **nodes**
Tracheobronchial nodes
Paratracheal nodes
R lymphatic or thoracic ducts

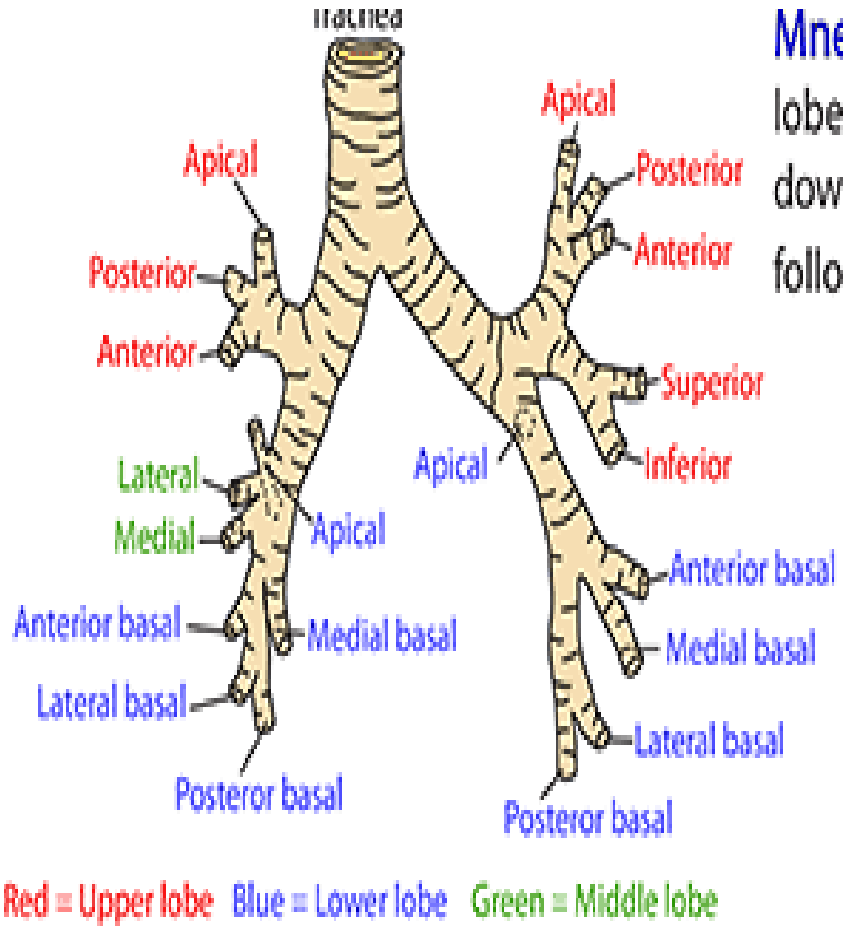
Nerve supply of the lungs

- By the pulmonary plexus in the root of each lung.
- The plexus has sympathetic (sympathetic trunk) and parasympathetic (vagus) nerves.



BRONCHOPULMONARY SEGMENTS

- Anatomic, functional and surgical unit of the lung.
- Pyramid shape with its apex towards the lung root
- Surrounded by CT
- Veins lie here
- Diseased segment can be removed surgically
- 10 segments in each lung



RIGHT LUNG

- SUPERIOR LOBE (3)

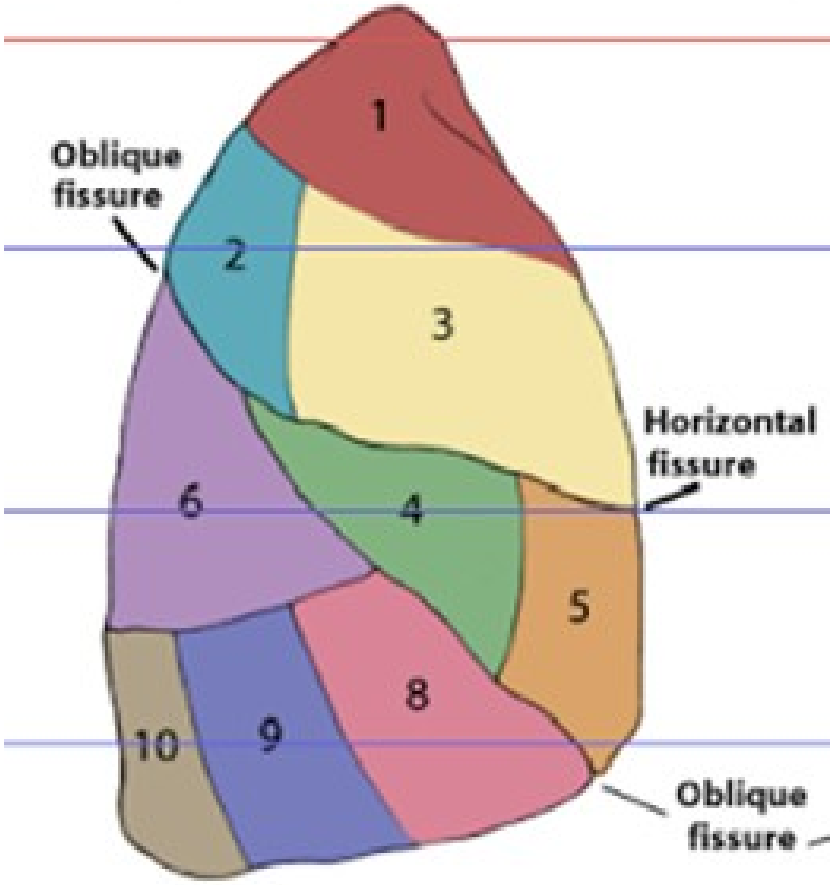
1. APICAL
2. POSTERIOR
3. ANTERIOR

- MIDDLE LOBE (2)

4. LATERAL
5. MEDIAL

- INFERIOR LOBE (5)

6. SUPERIOR(APICAL)
7. MEDIAL BASAL
8. ANTERIOR BASAL
9. LATERAL BASAL
10. POSTERIOR BASAL



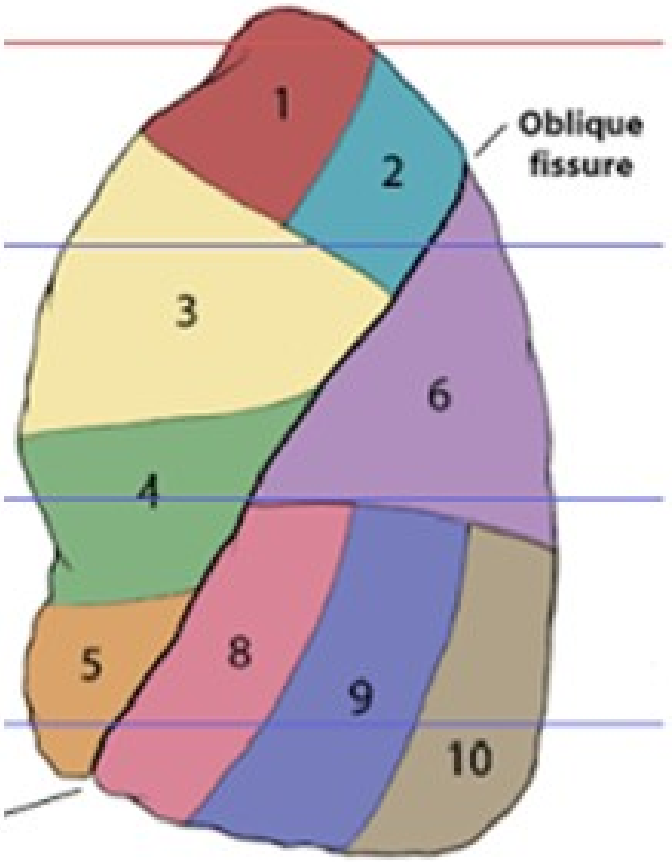
LEFT LUNG

- SUPERIOR LOBE (5)

1. APICAL
2. POSTERIOR
3. ANTERIOR
4. SUPERIOR LINGULAR
5. INFERIOR LINGULAR

- INFERIOR LOBE (5)

6. SUPERIOR(APICAL)
7. MEDIAL BASAL
8. ANTERIOR BASAL
9. LATERAL BASAL
10. POSTERIOR BASAL



THANK YOU